

ST03 ion-channel counterscreen, third chemistry shipment

Final Report

Study BW17_0167 (LIMS B-ACC-2017-02854)

Client: Selverant Therapeutics Inc

Study lead: Dr B. Rowcroft, Screening Operations Director

Report date: 14 June 2017

1 Summary

40 compounds from the Selverant aminoquinazoline series 03 were tested for Selverant Therapeutics Inc in the hERG automated patch clamp. Testing started on 25 May 2017 and results were reported on 14 June 2017. All accepted runs met plate acceptance criteria. 2 of 40 compounds had a hERG IC₅₀ below 1 μ M. Most compounds (37 of 40) inhibited hERG with IC₅₀ below 10 μ M, so margins will depend on free therapeutic exposure.

Table 1 Study summary

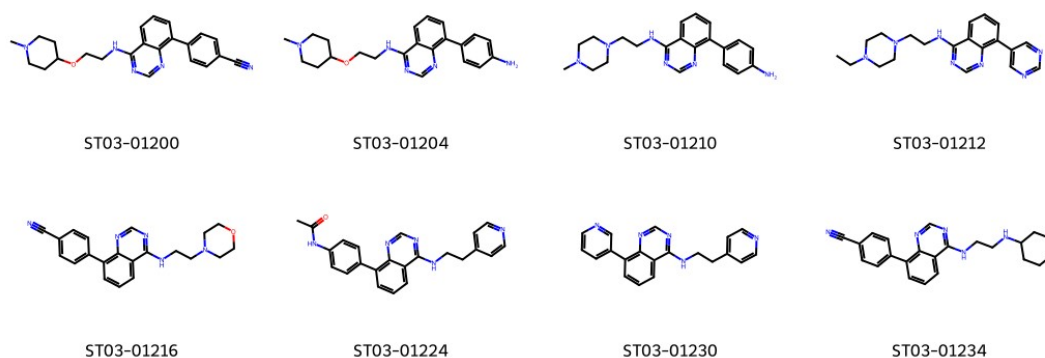
Metric	Value
Project	BW17_0167
Client	Selverant Therapeutics Inc
Compounds tested	40
Series / chemotype	Selverant aminoquinazoline series 03 (Aminoquinazoline)
Assays	hERG patch clamp
Turnaround (calendar days)	20
Price	GBP 10,900
hERG IC ₅₀ <1 μ M	2
hERG IC ₅₀ 1-10 μ M	35
hERG IC ₅₀ $\geq$ 10 μ M	3
hERG median IC ₅₀ (μ M)	3.7
E-4031 control IC ₅₀ (nM)	16

2 Objective

To rank the screening set for hERG automated patch clamp liability so that Selverant Therapeutics Inc could choose analogues for progression.

3 Test items

40 compounds were received as 10 mM DMSO solutions in one test lot on 25 May 2017 and logged against B-ACC-2017-02854. Identity and concentration were taken as supplied by the client. Stocks were stored at -20 °C and thawed once for testing.



4 Methods

hERG automated patch clamp

HEK293 cells stably expressing hERG (Kv11.1) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells are held at -80 mV, depolarised to +40 mV for 2 s and repolarised to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final). Each concentration is held until the tail-current block reaches a steady state (change below 5% over the last three sweeps), with a minimum of 3 and a maximum of 8 minutes; compounds still drifting at 8 minutes are flagged as slow-onset.

E-4031 is run as a separate 6-point concentration response (1-300 nM) on every run.

- Plate acceptance: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- E-4031 IC₅₀ within 5-50 nM; runs outside this range are rejected and repeated
- IC₅₀ fitted per cell (four-parameter logistic); the compound IC₅₀ is the geometric mean of finite cell IC₅₀s when these are the majority of accepted cells, otherwise reported as >30 μ M
- Where precipitation is seen at 30 μ M, that concentration is excluded and the highest evaluable concentration is reported

5 Results

hERG results

All 40 compounds were tested. 2 of 40 compounds had a hERG IC₅₀ below 1 μ M. Most compounds (37 of 40) inhibited hERG with IC₅₀ below 10 μ M, so margins will depend on free therapeutic exposure. The median IC₅₀ was 3.7 μ M (2 below 1 μ M, 35 between 1 and 10 μ M, 3 at or above 10 μ M, of which 0 above the highest evaluable concentration). E-4031 gave an IC₅₀ of 16 nM in the accepted runs.

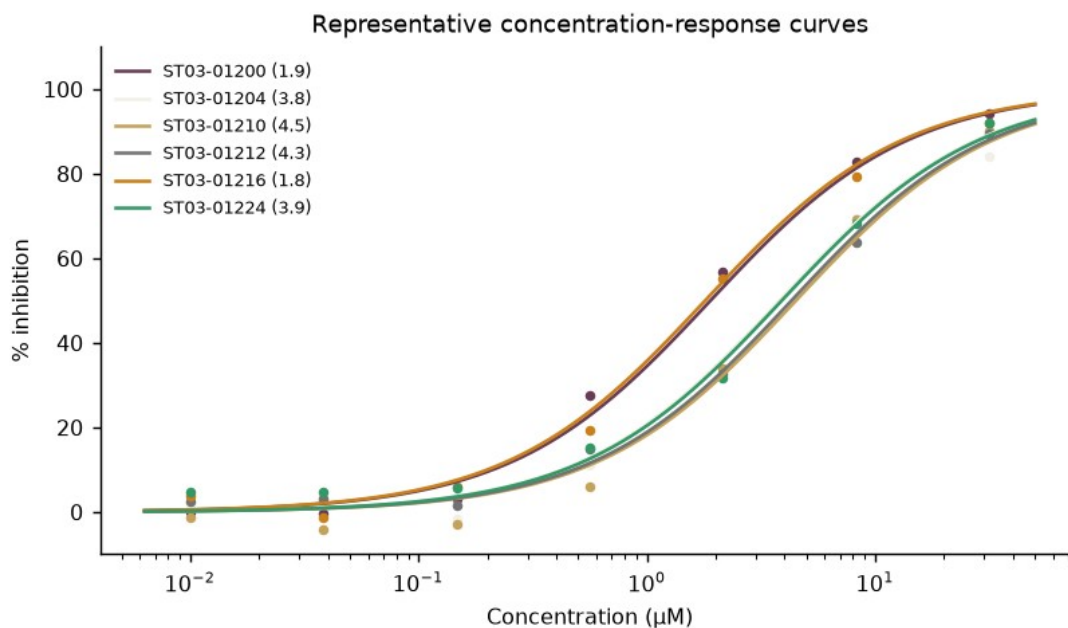
Table 2 hERG IC₅₀ by compound (μ M)

Compound	IC ₅₀ cell 1 (μ M)	IC ₅₀ cell 2 (μ M)	IC ₅₀ cell 3 (μ M)	IC ₅₀ geomean (μ M)	% inhibition @10 μ M	n cells (finite)	QC note
ST03-01200	1.9	1.2		1.5	90	2 (2)	
ST03-01204	3.8	6.2		4.9	74	2 (2)	

ST03-01210	4.5	4.7		4.6	72	2 (2)	
ST03-01212	4.3	3.4		3.9	72	2 (2)	
ST03-01216	1.8	1.2		1.5	88	2 (2)	
ST03-01224	3.9	2.3		3.0	83	2 (2)	
ST03-01230	15	7.6		11	46	2 (2)	
ST03-01234	4.3	3.8		4.0	64	2 (2)	
ST03-01236	6.3	8.0		7.1	55	2 (2)	
ST03-01242	2.0	1.1		1.5	80	2 (2)	
ST03-01248	2.8	3.5		3.1	74	2 (2)	
ST03-01254	4.9	4.0		4.5	70	2 (2)	
ST03-01258	8.1	8.3		8.2	59	2 (2)	
ST03-01264	1.4	2.4	3.2	2.2	86	3 (3)	
ST03-01270	4.9	5.0		4.9	61	2 (2)	
ST03-01276	6.6	4.3		5.3	59	2 (2)	
ST03-01284	3.2	2.6		2.9	76	2 (2)	
ST03-01288	3.5	3.6		3.6	81	2 (2)	
ST03-01290	3.5	3.7		3.6	73	2 (2)	
ST03-01296	7.5	15		11	50	2 (2)	
ST03-01300	6.3	2.2		3.7	79	2 (2)	
ST03-01302	4.2	3.6		3.9	72	2 (2)	
ST03-01306	4.4	4.0	3.3	3.9	70	3 (3)	
ST03-01308	3.3	5.4		4.2	79	2 (2)	
ST03-01312	2.8	2.7		2.7	83	2 (2)	
ST03-01314	1.1	0.80		0.95	85	2 (2)	
ST03-01318	0.57	0.40		0.48	100	2 (2)	
ST03-01326	2.1	1.6		1.8	92	2 (2)	
ST03-01330	5.2	3.7		4.4	66	2 (2)	
ST03-01332	1.7	1.7		1.7	87	2 (2)	
ST03-01336	1.7	1.1		1.4	81	2 (2)	
ST03-01338	7.4	3.0	4.9	4.8	72	3 (3)	
ST03-01342	6.4	3.5		4.7	65	2 (2)	
ST03-01344	1.9	3.0	2.4	2.4	80	3 (3)	
ST03-01356	1.1	2.3		1.6	85	2 (2)	
ST03-01360	3.0	4.2	2.2	3.0	82	3 (3)	
ST03-01372	8.9	12		10	50	2 (2)	
ST03-01380	4.3	3.1		3.7	79	2 (2)	
ST03-01384	2.9	4.0		3.4	76	2 (2)	
ST03-01386	4.2	4.7		4.4	70	2 (2)	

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 16 nM (acceptance 5-50 nM)

Compound IC₅₀ = geometric mean of finite cell IC₅₀s when they are the majority of accepted cells; otherwise reported as >30 μ M (or >10 μ M where solubility-limited)



6 Deviations

No deviations from the study plan.

7 Discussion

2 of 40 compounds had a hERG IC₅₀ below 1 µM. Most compounds (37 of 40) inhibited hERG with IC₅₀ below 10 µM, so margins will depend on free therapeutic exposure. These are screening data; mechanism statements are outside the scope of this study.

8 Conclusion

Recommended next steps: prioritise compounds with hERG IC₅₀ at or above 10 µM; for any compound below 10 µM, compare the IC₅₀ with projected free plasma exposure before progression.

9 Records

Raw data, run sheets and instrument files are retained for 5 years under B-ACC-2017-02854.

Name	Role	Signature	Date
Bethany Rowcroft	Screening Operations Director	<u>B Rowcroft</u>	14 June 2017
Ines Alvarado	Assay Quality Coordinator	<u>I Alvarado</u>	13 June 2017